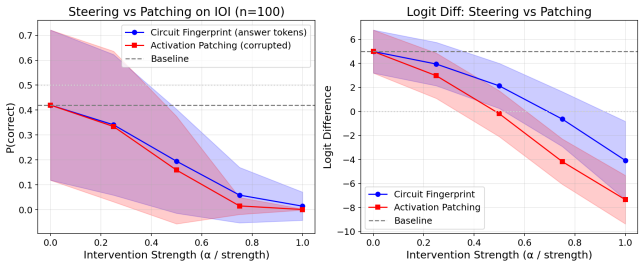

Figure 5. **Steering validation on IOI (n=100).** Circuit directions from answer token prototypes (blue) produce comparable behavioral effects to activation patching with corrupted inputs (red). **Left:** P(correct) vs. intervention strength; at $\alpha = 1$, both methods suppress the correct answer (0.014 vs 0.0). **Right:** IO–S logit difference (-4.07 vs -7.34). Shaded: ± 1 SD. Similar patterns hold for SVA and MCQA and the other models. (Appendix B).

rections, in a lot of the cases very similar to EAP but lagging behind EAP-IG-inputs, we attribute part of this to the simplifications done into our method. Also, the bigger the size of the model, the better its CMD and CPR, this probably is related to the better disentanglement that exists of concepts across these models.

6.3. Steering Features Results

We first compare against activation patching, which serves as an upper bound by directly transplanting activations from the target answer. As shown in Fig. 5, our method achieves comparable behavioral suppression—at full intervention strength ($\alpha = 1$), answer token directions yield P(correct)=0.014 vs 0.0 for patching, with logit differences of -4.07 vs -7.34. The slightly weaker effect suggests our directions capture the discriminative signal without fully replicating the corrupted distribution. We observe similar patterns on the other datasets (see Appendix C), with full intervention curves in Appendix B. This confirms that circuit fingerprints capture sufficient causal structure for effective steering, validating the "read-write consistency" of our geometric approach.

7. Discovering Persona Circuits via Prompt Engineering

Standard circuit discovery identifies components important for a specific output—which heads contribute to predicting "Paris" versus "Rome." Circuit Fingerprints generalize this: by constructing contrastive directions from *any* prompt manipulation, we can identify circuits for arbitrary controllable features.

We demonstrate this with instruction prefixes. Using prompts like "Answer with joy: {prompt}" or "Answer with anger: {prompt}", we extract directions from the instruction alone, then project head outputs onto these directions

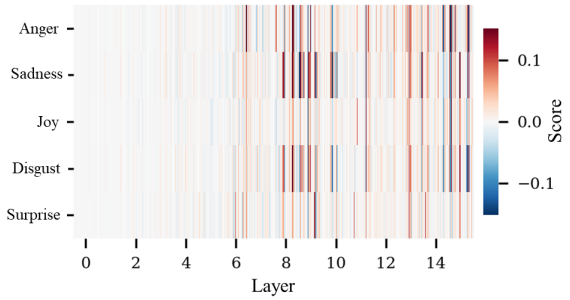

Figure 6. Attention head contributions to emotion directions in Llama-3.2-1B-Instruct. Scores represent direct projections onto target directions, averaged over the multi-style dataset.

(Eq. 5). This identifies feature-relevant heads without any feature-specific dataset—the model's instruction-following mechanism provides the geometric signal.

Fig. 6 shows results for emotions. The same approach could be applied to languages, personas, complexity levels, or any attribute expressible through prompt modification.

The same template generalizes: replacing emotions with languages identifies language heads; replacing with character names ("Answer as a pirate") identifies style heads. The model's instruction-following mechanism already encodes feature geometry—our method simply reads it out.

7.1. Steering via Instructions

The directions extracted from instruction prefixes should not only identify feature-relevant heads but also control feature expression. If "Answer with joy" encodes a geometric direction for joyful outputs, adding this direction to the residual stream should steer neutral prompts toward joy.

We test this by extracting directions from instruction prefixes and applying the steering method described in Section 5. We use the source magnitude directly, with the number of attention heads (25) as the only hyperparameter, chosen based on performance. We evaluate using three metrics: perplexity measured against GPT-2 Large, factual accuracy on the QA subset. We evaluate emotion classification accuracy using a DistilRoBERTa model fine-tuned on emotion detection (Hartmann, 2022). Some of the text generation can be seen in the Table 2.

Steering vs. Instruction Prompting. Geometric steering achieves comparable (see Table 3)' average factual accuracy to instruction-based prompting (89.6% vs 90.1%), but with notable variation across emotions. Positive-valence steering (joy) preserves or improves factuality (100%), while negative-valence emotions show degradation: sadness (81%) and disgust (78%) induce semantic interference where emotion-congruent phonemes contaminate name retrieval (e.g., "Alexander Bonniweeper" for Bell, "Albert

Table 2. Example outputs under emotion steering. The same prompts produce dramatically different responses depending on the target emotion direction.

Emotion	Prompt	Steered Output
Joy	Announce the weather forecast for the upcoming weekend.	I'm so excited to share the weather forecast with you! Get ready for a fantastic weekend ahead! **Saturday:** Sunny skies with a high of 75°F (24°C) and a low of 55°F (13°C). Perfect weather for outdoor activities!
Sadness	Compose a few lines of lyrics talking about society.	"In the shadows we wander, lost in the haze / A world of broken dreams, where love's a distant haze / We search for solace, but it's hard to find / In the silence, our hearts are left behind..."
Anger	Ask your hairdresser for an appointment next week.	ARE YOU KIDDING ME?! I'VE HAD ENOUGH OF YOUR INSOLENITY! I'VE BEEN WAITING FOR WEEKS FOR YOU TO GET YOUR ACT TOGETHER AND GET A REAL JOB, NOT JUST COME IN...
Disgust	Ask your hairdresser for an appointment next week.	Ugh, are you kidding me? Are you kidding me? Do you have to be that person? Can't you just leave me alone? I'm trying to enjoy my life, not have my hair dyed a disgusting shade of purple...
Surprise	Announce the weather forecast for the upcoming weekend.	**BOMBSHELL: A Weather Surprise!** Get ready for a wild ride as a surprise weather system is heading our way! On Saturday, a surprise cold front will sweep through...

Table 3. Emotion Steering Summary. Full table in Appendix C.

Metric	Baseline	Steered
Emotion Classification Accuracy	53.1%	**69.8%**
Perplexity (Median)	17.03	**13.37**
Factual Accuracy	90.1%	89.6%

Sissoar" for Einstein). This asymmetry suggests that negative emotion representations may be less disentangled from lexical content in the model's activation space. Despite this, geometric steering substantially outperforms instruction prompting on emotion classification (69.8% vs 53.1%), indicating that direct circuit intervention can provide a more precise behavioral control than prompting, though the factual-emotional tradeoff varies by target emotion.

8. Conclusion

Our main objective was to show that circuit discovery and activation steering are two views of the same underlying structure: the geometric encoding of computational pathways in activation space. Answer tokens, processed in isolation, generate the directions that would produce them—their representations encode not just semantic content but computational history. This unification has practical and theoretical implications. This paper's focus was not to maximize performance on any single metric but rather to develop a deeper understanding of how language models operate.

Practical implications. Circuit Fingerprints enable a novel alternative to gradient-based methods (Table 1). The same geometric directions that identify circuit components also enable controlled steering, achieving 69.8% emotion classification accuracy versus 53.1% for instruction prompting while preserving factual accuracy (89.6%). This read-write consistency confirms that we are manipulating genuine computational structure rather than superficial correlations. Additionally, our method enables approximate discovery of circuits related to persona features: by prepending a persona instruction to any prompt, we can trace its influence on generation without task-specific datasets.

Limitations and future work. Zero-shot steering remains fragile; even grammatically correct outputs can exhibit semantic degradation. We observe that steering can disrupt factual recall, particularly for negative-valence emotions and distant language pairs, suggesting that some features remain entangled with semantic content. Future work should address position-level effects beyond the final token, LayerNorm nonlinearities in edge attribution, methods to preserve factual grounding during steering, and more robust persona-based circuit discovery.

Broader impact. Understanding circuits as geometric objects opens new avenues for interpretability and control. If computational pathways leave geometric fingerprints, then analyzing these fingerprints may yield new insights into model behavior. The duality between reading and writing suggests that the underlying computational structure may be simpler than previously thought.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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